Lab Task 4

Problem Scenario

A healthcare institution aims to develop a reliable breast cancer classification system using different Ensemble Learning techniques. The system will analyze tumor characteristics and classify tumors as Benign or Malignant. Instead of relying on a single Decision Tree, the institution wants to investigate different ensemble approaches, including Bagging, Boosting, Stacking, and Random Forest, to improve classification performance and reliability.

Tasks

1. Download the Breast Cancer Wisconsin Diagnostic dataset and load it using pandas.
2. Check for missing values in each column. Handle missing values appropriately and justify the selected method.
3. Identify categorical and numerical features.
4. Perform necessary data preprocessing, including feature encoding and feature scaling (if required).
5. Split the dataset into training (80%) and testing (20%) sets.
6. Create and train the following Ensemble Learning classifiers:
    - **Bagging Classifier:** Use Decision Tree as the base estimator and train multiple trees using bootstrap samples.
    - **Boosting Classifier:** Implement AdaBoost or Gradient Boosting to sequentially train weak learners.
    - **Stacking Classifier:** Combine multiple base classifiers (e.g., Decision Tree, K-Nearest Neighbors, and Support Vector Machine) using a Logistic Regression meta-classifier.
    - **Random Forest Classifier:** Train multiple Decision Trees using bootstrap samples and random feature selection.
7. Analyze the model structure and characteristics of each ensemble technique:
    - Analyze the number of estimators and their contribution.
    - Visualize or analyze individual Decision Trees where applicable.
    - Analyze feature importance for applicable models.

    - Examine the base learners and meta-classifier used in Stacking.
8. Make predictions on the test dataset using all four classifiers.
9. Generate confusion matrices and calculate the following evaluation metrics for each model:
    - Accuracy
    - Precision
    - Recall
    - F1-score
10. Compare the performance of Bagging, Boosting, Stacking, and Random Forest classifiers using a comparative table and graphical visualization.
11. Identify the best-performing model based on the evaluation results and justify your selection.